FTIN: Frequency-Time Integration Network for Inertial Odometry

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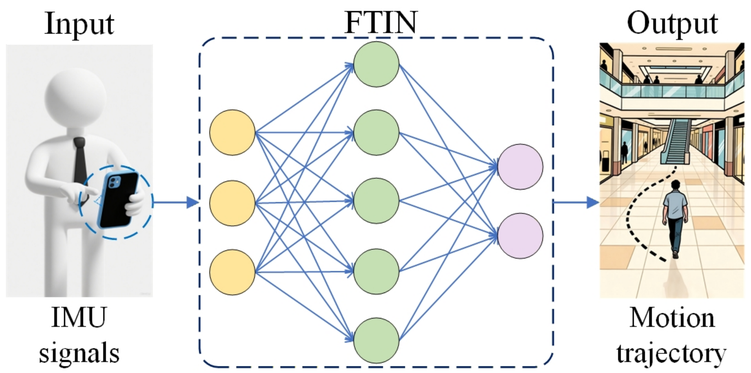
Motivations

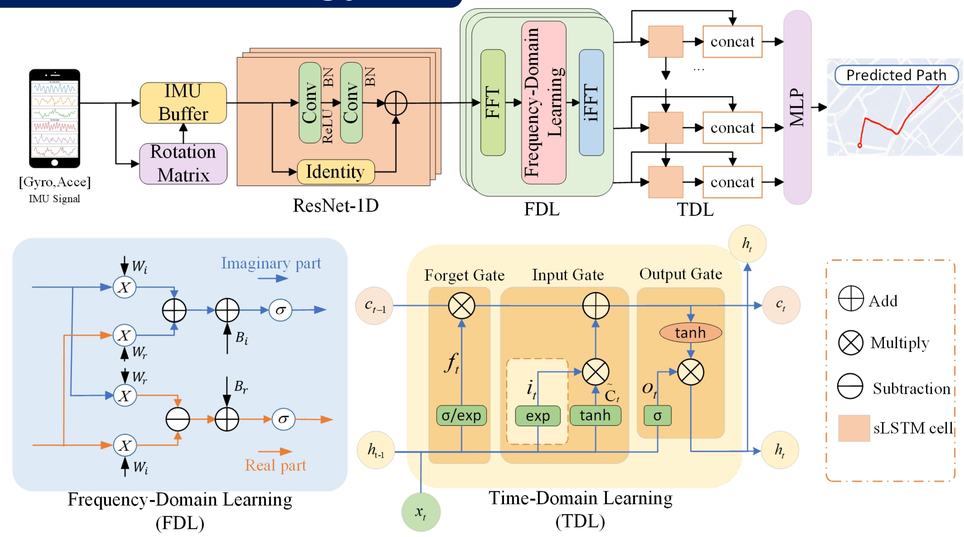
Inertial odometry (IO) leverages inertial measurement unit (IMU) signals for cost-effective localization. However, high IMU sampling rates introduce substantial redundancy that impedes IO's ability to attend to salient components, thereby creating an information bottleneck.

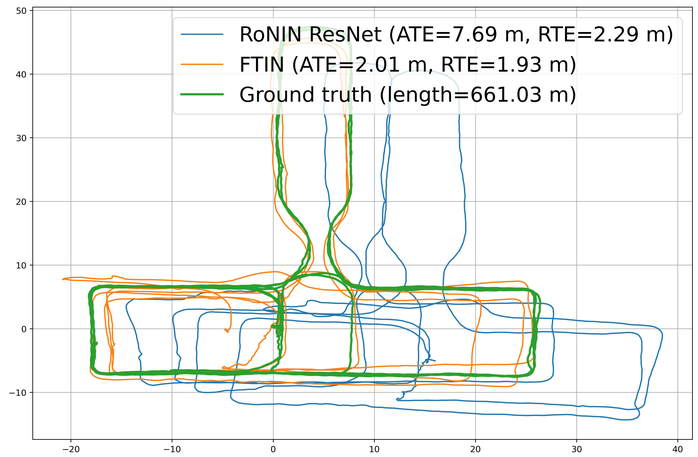
Key Contributions

- Reinterpret IMU signals from a frequency-domain perspective to reduce redundancy and enable contextual motion modeling;
- Enhance the ResNet architecture with frequency-domain learning to integrate local motion features and global contextual dependencies;
- Introduce a Scalar LSTM module to improve time-domain contextual motion modeling and facilitate effective cross-domain feature fusion;



Methodology





Representative trajectory from the RoNIN dataset

Experimental Results

Table 1: Overall Trajectory Prediction Accuracy and Ablation Study. All metrics are in meters. Best results are in **bold**.

Datasets	Metric	RoNIN ResNet	RoNIN LSTM	RoNIN TCN	RNIN	IMUNet	TLIO	FTIN	FTIN vs RoNIN ResNet	Ablation Study		
										+FDL	+TDL	+FDL&TDL
RoNIN	ATE	6.799	6.136	5.500	5.686	5.550	4.417	**3.875**	43.0%	5.877	4.690	**3.875**
	RTE	3.523	3.595	3.563	3.449	3.387	3.507	**3.061**	13.1%	3.323	3.223	**3.061**
RIDI	ATE	2.608	3.294	2.743	1.851	2.645	2.641	**1.727**	33.8%	1.962	2.367	**1.727**
	RTE	2.881	3.136	2.476	2.040	2.788	3.105	**1.981**	31.2%	2.194	2.560	**1.981**
RNIN	ATE	1.182	1.893	1.561	1.603	1.417	1.493	**1.096**	7.3%	1.193	1.097	**1.096**
	RTE	2.036	2.732	2.108	1.842	1.872	1.728	**1.114**	45.3%	1.199	1.123	**1.114**
TLIO	ATE	1.445	2.087	4.682	1.611	**1.432**	1.779	1.479	-2.4%	**1.383**	1.448	1.479
	RTE	3.758	5.990	14.524	4.637	**3.474**	5.127	3.552	5.5%	3.595	3.707	**3.552**
OxIOD	ATE	3.557	4.277	26.442	4.343	9.822	5.071	**3.446**	3.1%	3.518	3.452	**3.446**
	RTE	1.324	1.683	5.628	1.340	2.307	1.445	**1.165**	12.0%	1.308	**1.165**	**1.165**
IMUNet	ATE	18.040	10.182	38.272	16.552	11.847	17.190	**10.171**	43.6%	14.610	17.557	**10.171**
	RTE	9.251	7.382	29.620	8.566	7.792	9.312	**7.289**	21.2%	8.598	7.809	**7.289**
Parameters (MB)		4.635	13.589	13.916	5.259	**3.659**	5.424	4.474	-	-	-	-

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